

Студент: Лиунша Арсений Максимович

Задание А

А. 1_1. Пирамидки чисел

Язык: Python

Файл: A-133043695-python-lyceum-OK_50.py

Исходный код:

```
BYBWRW
```

Итоговый балл: 50

Задание В

В. 1_2. Квантовые драконы

Язык: Python

Файл: В-133043189-python-lyceum-OK_50.py

Исходный код:

```
77812273803
```

Итоговый балл: 50

Задание С

С. 2_1. Трибуна

Язык: Python

Файл: C-133037095-pypy3_7_1_0-WrongAnswer_40.py

Исходный код:

```
s = int(input())
x = int(input())

sq = int(s**0.5)

print((sq+1)*2 + (sq-1)*2)
```

Итоговый балл: 40

Задание D

D. 2_2. Идеальная команда

Язык: Python

Файл: D-133038690-пуру3_7_1_0-ОК_100.py

Исходный код:

```
n = int(input())

ans = 0
sieve = [False, False] + [True] * (max(n - 1, 0))

for i in range(2, n+1):
    if not sieve[i]:
        continue

    for j in range(i + i, n+1, i):
        sieve[j] = False

print(sum(sieve) + 1)
```

Итоговый балл: 100

Задание F

F. 3_1. Драгоценная опасность

Язык: Python

Файл: F-133036057-python-lyceum-OK_100.py

Исходный код:

```
select t.year, round(sum(cast(t.quantity as real) / cast(c.danger as real)), 2)
from transactions as t

join owners as ow on t.owner_code = ow.code
join operations as op on t.operation_code = op.code
join wealth as w on t.wealth_code = w.code
join creatures as c on ow.view_code = c.code

where op.operation = 'purchase'
      and w.variety in ('jewelry', 'pearls', 'diamond')
      and ((t.year % 2) = 1)

group by t.year

order by t.year asc
```

Итоговый балл: 100

Задание G

G. 3_2. Гоблинский банк

Язык: Python

Файл: G-133040304-python-lyceum-OK_100.py

Исходный код:

2639

Итоговый балл: 100

Задание Н

Н. 3_3. Уплата налогов

Язык: Python

Файл: Н-133044757-python3_docker-OK_100.py

Исходный код:

```
from os import listdir, system
import json
import csv
import zipfile

fname = input()
who = input()
year = input()
tax = int(input())

zip = zipfile.ZipFile(fname)
zip.extractall()
zip.close()

total = 0

stats = []

for cave in listdir("./bank"):
    for safe in listdir(f"./bank/{cave}"):
        try:
            with open(f"./bank/{cave}/{safe}/report.json") as f:
                report = json.load(f)
        except FileNotFoundError:
            continue

        if report["owner"] != who:
            continue

        if year not in listdir(f"./bank/{cave}/{safe}/"):
            continue

        outcome = 0
        for trans in listdir(f"./bank/{cave}/{safe}/{year}"):
            with open(f"./bank/{cave}/{safe}/{year}/{trans}") as f:
                data = list(csv.DictReader(f))
            for entry in data:
                if entry["expense"] == "sale":
                    outcome += int(entry["outcome"])
            stats.append((outcome, int(report["benefits"])))
    try:
```

```
        system("rm -r bank")
    except:
        pass

    # print(stats)

    ans = sum([i[0] for i in stats])
    if stats:
        benefit = stats[0][1]
        tax = max(tax - benefit, 5)

    ans = int(ans * tax / 100)

    print(ans)

    # print(int(total * tax / 100))
```

Итоговый балл: 100

Задание I

I. 4_1. Коллоквиум

Язык: Python

Файл: I-133049156-python-lyceum-OK_100.py

Исходный код:

```
from fastapi import FastAPI, Query, Response
import uvicorn
import datetime
import json

dates = []
with open("input.txt") as f:
    for s in f:
        data = json.loads(s)
        date_start = datetime.datetime.strptime(data["date"] + " " + data["time"], "%d.%m.%Y %H:%M:%S")
        date_end = date_start + datetime.timedelta(minutes=data["duration"])
        dates.append((date_start, date_end))

print(dates)

app = FastAPI()

@app.get("/find")
async def handle(date: str):
    d, m, y = map(int, date.split("."))
    d_s = datetime.datetime(day=d, month=m, year=y, tzinfo=datetime.UTC)

    d_s_fwd = d_s
    while True:
        d_e = d_s_fwd + datetime.timedelta(hours=23, minutes=59)
        if all([max(d_s_fwd, d2_s) > min(d_e, d2_e) for d2_s, d2_e in dates]):
            break
        d_s_fwd += datetime.timedelta(days=1)

    d_s_back = d_s
    while True:
        d_e = d_s_back + datetime.timedelta(hours=23, minutes=59)
        if all([max(d_s_back, d2_s) > min(d_e, d2_e) for d2_s, d2_e in dates]):
            break
        d_s_back -= datetime.timedelta(days=1)

    if d_s_fwd - d_s <= d_s - d_s_back:
        res = d_s_fwd
    else:
        res = d_s_back
```

```
        return Response(res.strftime("%d.%m.%Y"))

if __name__ == '__main__':
    uvicorn.run(app, host='0.0.0.0', port=8080)
```

Итоговый балл: 100

Задание J

J. 4_2. Молчащая мандрагора

Язык: Python

Файл: J-133051148-python-lyceum-WrongAnswer_0.py

Исходный код:

```
from fastapi import FastAPI, Query, Response
import uvicorn
import datetime
import json

data = {}

def needs_pour(plant, date):
    interval, maxdelay, date_add, pours = data[plant]

    if date <= date_add:
        return False

    delay = maxdelay
    cur = date_add
    while cur != date:
        if cur in pours:
            if (cur - date_add).days % interval != 0:
                return False

            delay = maxdelay

        if delay < 0:
            return False

        cur += datetime.timedelta(days=1)
        delay -= 1

    return (date - date_add).days % interval != 0

app = FastAPI()

@app.get("/add/{plant}/{interval}/{maxdelay}")
async def handle(plant: str, interval: str, maxdelay: str, date: str):
    d, m, y = map(int, date.split("."))
    d_s = datetime.datetime(day=d, month=m, year=y, tzinfo=datetime.UTC)

    data[plant] = (int(interval), int(maxdelay), d_s, set())
    return Response()

@app.get("/task/")
```

```

async def handle2(date: str):
    d, m, y = map(int, date.split("."))
    d_s = datetime.datetime(day=d, month=m, year=y, tzinfo=datetime.UTC)
    a = []
    for plant in data.keys():
        if needs_pour(plant, d_s):
            a.append(plant)
    a.sort()
    return Response(', '.join(a))

@app.get("/watering/{plant}/")
async def handle3(plant: str, date: str):
    d, m, y = map(int, date.split("."))
    d_s = datetime.datetime(day=d, month=m, year=y, tzinfo=datetime.UTC)

    data[plant][3].add(d_s)
    return Response()

if __name__ == '__main__':
    uvicorn.run(app, host='0.0.0.0', port=8080)

```

Итоговый балл: 0